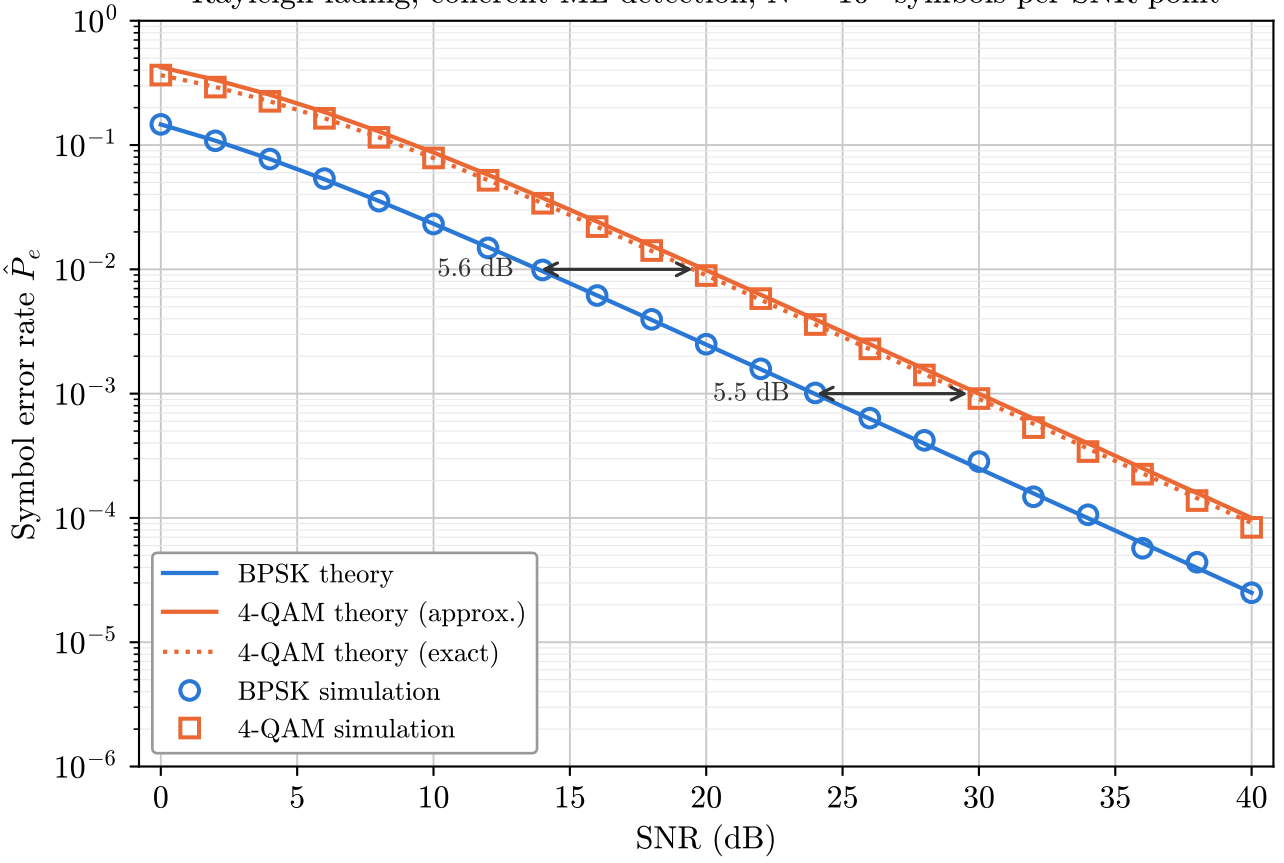
Rayleigh fading, coherent ML detection, N = $10^6$ symbols per SNR point
Symbol error rate $\hat{P}_e$
SNR (dB)
$10^0$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
0
5
10
15
20
25
30
35
40
5.6 dB
5.5 dB
BPSK theory
4-QAM theory (approx.)
4-QAM theory (exact)
BPSK simulation
4-QAM simulation